



BUYERS' GUIDE

number of smartphones with great hardware configuration.

Image quality is everything

Console based or PC VR headsets mean more than simply strapping smartphones to your head with two convex lenses inside the cardboard or plastic casing. These provide you with graphic-intense immersion. Advancements in room-scale motion controllers, sensors, tracking and built-in cameras have made VR more interactive and has made viewing video in high definition, while moving around, a lot of fun.

VR could also be used in training and simulation in fields like education, medicine, surgery, construction and armed forces using VR headsets. This could save costs while effectively delivering content to consumers.

Built-in headphones. There is something called head-related transfer function (HRTF) that is being incorporated into VR sets. You have two ears, and to experience a simulated environment it is important you know the source of sound that is coming from various directions, and how it is localised through diffraction and reflection. An alternative to this can be noise-cancellation headphones, but these do not do what HRTF does.

Built-in camera for 360-degree view. In VR, the environment is controlled by you, but in 360, all you get to do is go along with the visuals. This aspect may confuse some VR enthusiasts and trick them into buying headsets with 360-degree views.

Motion controller and game pad support. The thing that really sets the high-end VR headsets apart is their motion controllers. These let you interact with the 3D environment, while tracking your movements and gestures with advanced capacitive sensors and tracking cameras.

If you are a gamer, you would be familiar with guns or virtual wands. These work on the same principles of motion sensing. Smartphones have inbuilt accelerometers and gyro sensors to provide the same effect, but this would only be a fraction of visual immersions that VR consoles provide.

Refresh rate. To experience real VR, the headset must have a refresh rate of 90Hz to 120Hz (cinema mode), whereas a smartphone can have a refresh rate of 60Hz.

Wireless. Though smartphones have a commendable battery life, a wireless VR headset can give you unrestricted movement with comfort. Some prototypes give about 90 minutes of VR view.

Sensors. Accelerometers, gyroscopes, magnetometers, external constellation-tracking cameras and eye-tracking systems are some common sensors loaded onto a VR console. Sensors bring sophistication in terms of tracking movements and helping you coordinate with the virtual world.

The best VR consoles in market

Built-in headphones and mic in Oculus Rift. Headphones work best in a sound-proof room but HRTF helps in rendering impressive sound with respect to VR that Oculus Rift presents you with.

Precision tracking of movement with HTC Vive. With HTC Vive you can experience precise movements that help in artificial locomotion. Using the front camera, walking around in room-size tracked spaces becomes even more realistic.

The best gaming VR money can buy is with PlayStation VR. This console is best suited for gaming. It may require you to buy a PlayStation camera, but it is affordable in its range. Its GPU enhances the output for a more realistic gaming experience. **RFY**

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